

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/03

Applications Paper and Planning Question

15 September 2014

Paper 3

2 hours

Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This document consists of **19** printed pages.

[Turn over]

Answer **all** the questions in this section.

- 1 Restriction enzymes are used during genetic engineering. Fig 1.1 shows restriction sites for *EcoRI*, *BamHI* and *AluI*.

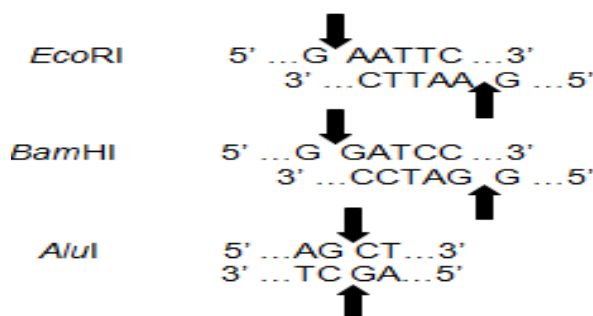


Fig. 1.1

- (a) Describe the natural function of these restriction enzymes. [3]

1. Recognize, bind and cut viral genome/foreign DNA that entered the bacterial cell;;
2. By hydrolyzing phosphodiester bonds between adjacent nucleotides;;
3. Protecting the bacterium from attack by viruses/bacteriophages;;

- (b) Explain why it is more suitable to use *EcoRI* than *AluI* in genetic engineering. [3]

1. *EcoRI* produces restriction fragments with sticky ends, allowing cut plasmid and cut gene of interest/pieces of DNA to anneal via complementary base pairing (and hydrogen bond formation);;
2. *AluI* produces restriction fragments with blunt ends, hence requires addition of linkers;;
3. which would need to be recognized, bind and cut by another enzyme to produce restriction fragments with sticky ends before annealing with the cut gene of interest, hence process takes longer/require more energy;;

A bacterial plasmid is shown in Fig. 1.2. Apart from the DNA sequence shown, there are no target sequences for the three restriction enzymes in the rest of the plasmid.



Fig 1.2

(c) With reference to Fig. 1.1 and Fig. 1.2, state the number of fragments of DNA present after the plasmid has been cut with

- (i) *EcoRI* [1]
1;;
- (ii) *EcoRI* and *BamHI* [1]
2;;

Eukaryotic genes can be cloned using bacteria cells whereby the gene of interest is amplified by the bacteria as it replicates the plasmid before division. The Polymerase Chain Reaction (PCR) is another method for amplifying DNA and it is carried out *in vitro* using a thermal cycler.

The first stage of the PCR cycle involves heating the PCR mixture to cause DNA denaturation.

(d) Describe the next two stages of PCR.

- (i) 2nd Stage: [2]
 - 1. Annealing of DNA primers by cooling to 65°C;;
 - 2. Primers anneal via complementary base pairing at the 3' end of single stranded DNA;;
- (ii) 3rd stage: [2]
 - 1. primer extension by heating to 72°C;;
 - 2. Taq polymerase adds free nucleotides/dNTPs to 3' ends on both primers to synthesize rest of the fragment (new strands of DNA);;

[Total: 12]

- 2 Severe combined immunodeficiency (SCID) represents a group of rare and sometimes, fatal disorders characterized by failure of the immune system. All forms of SCID are inherited, with X-linked SCID being the most common.

(a) Describe the genetic basis of X-linked SCID that leads to failure of the immune system. [4]

1. Interleukin-2-Receptor Gamma (IL2RG) gene found on the X chromosome encodes common gamma chain subunit of the interleukin receptor;;
2. Mutations of IL2RG gene results in a non-functional version of the common gamma chain formation resulting in non-functional interleukin receptor
OR
no common gamma chain formation resulting in non-functional /no interleukin receptor;;
3. Communication between T and B cells/lymphocytes is not possible;;
4. Common gamma chain is shared by many interleukin receptors, mutations cause widespread defects in interleukin signalling;;
5. These interleukins and their receptors are also involved in the development and differentiation of T and B cells;; resulting in a complete failure of immune system.

Point 1 for full credit. Max 4

(b) The best treatment for SCID is a bone marrow transplant from an immunologically matched sibling. When no matched donor is available, unmatched donors, such as parents, are recruited; these transplants are only around 70% successful. Success of gene therapy may rival or better that seen in these unmatched donor situations.

(i) Explain what is meant by gene therapy. [2]

1. Gene therapy is a technique for introducing a copy of normal functional allele into target cells with non-functional gene using a viral or non-viral vector;;
2. The newly introduced normal functional allele is transcribed and translated for the synthesis of a normal functional protein /to restore target cell to normal state/correcting functions and altering the phenotype of the target cell;;

(ii) Explain why SCID is a suitable choice for gene therapy. [2]

1. SCID is a single gene disorder/ Mutation in 2 copies of IL2RG gene in females/ mutation in 1 copy of IL2RG gene in males / mutation in 2 copies of ADA gene gives rise to SCID;;
2. Only require one copy of dominant normal/functional allele is required to be inserted into target cell to restore normal phenotype;;

(c) In 2001, gene therapy aimed at treating X-linked SCID succeeded for 15 French baby boys. However, 3 of the little boys later developed cancer and died.

Suggest how gene therapy could have contributed to the development of cancer in the 3 boys. [2]

1. inability to control the specific site of gene insertion / random integration of gene / insertional mutagenesis;;
2. disrupt control elements such as activators or silencers;;
3. activation of oncogenes/mutation of proto-oncogene;;
4. inactivation of tumour suppressor gene;;
5. inactivation of a gene controlling apoptosis;;
6. ref. uncontrolled cell division/proliferation;;

- (d) Another form of characterized SCID is caused by mutation of adenosine deaminase (ADA) gene.

A polymorphic DNA segment, DXS86 is found to be tightly linked to the ADA gene. This segment exhibits restriction fragment length polymorphism (RFLP) that can be used as a marker for the genetic screening of ADA SCID.

Fig. 2.1 shows the restriction sites of *Bgl* II (indicated by *) on a segment of DXS86 and a probe that can be used to detect the polymorphism. Allele 2 is linked to the mutant ADA allele.

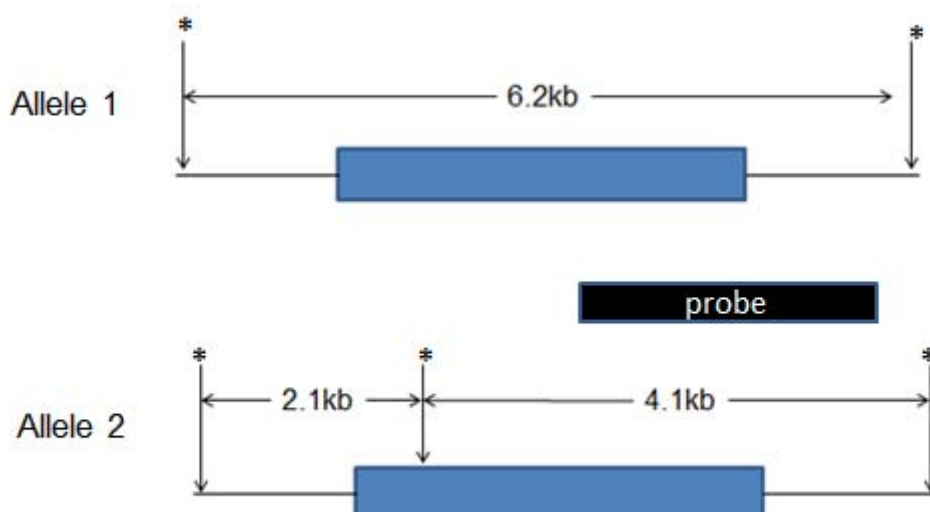


Fig.2.1

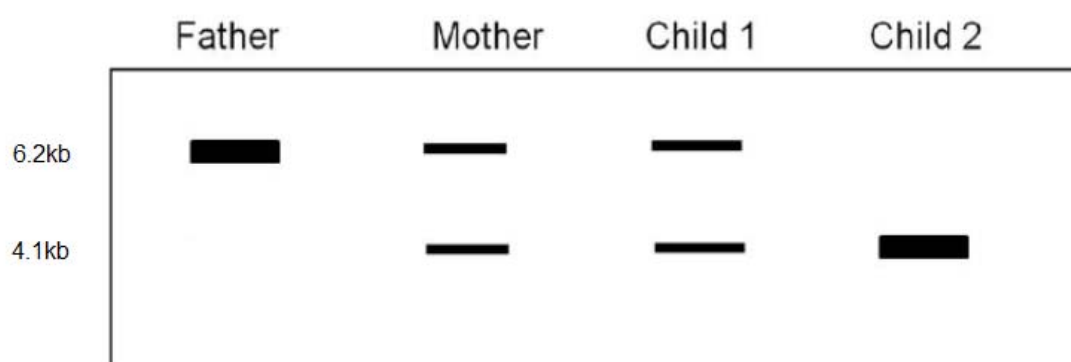


Fig. 2.2

Fig. 2.2 shows the genetic screening results for ADA SCID in a family. The amplified DNA containing DXS86 was digested with restriction enzymes followed by gel electrophoresis and Southern blotting using the probe shown in Fig. 2.1.

- (i) Based on Fig 2.1 and Fig. 2.2, explain if Child 2 is likely to get ADA SCID. [2]
 1. **Likely due presence of only one 4.1kb fragment implies that the child has 2 copies of mutant ADA alleles/homozygous recessive;**

(ii) If Child 1 shows symptoms of ADA-SCID and disregarding the possibility of crossing over between DXS86 and ADA loci during gamete formation, suggest a reason for the banding pattern obtained in Child 1. [2]

1. **Banding pattern of 4.1kb and 6.2 kb implies child 1 is a carrier / only has 1 mutant ADA allele/heterozygote;;**
2. **A gene mutation such as base-pair substitution/insertion/deletion has taken place within the restriction site of allele 2 such that the restriction site is no longer recognized and cut by restriction enzyme;;**
3. **As such, allele 2 is not cut and a 6.2kb fragment is generated and the mutant ADA allele is now perceived to be absent;;**

[Total: 14]

- 3 Rice, *Oryza sativa*, is the staple diet in many parts of the world. It lacks a number of important nutrients, including β carotene, from which vitamin A is synthesised. Adequate concentrations of vitamin A give protection from night blindness. Higher concentrations act as an antioxidant that may give some protection from cancer and heart disease.

Golden rice, which contains β carotene, was developed by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium.

The expression of these inserted genes is under the control of promoters found only in the endosperm (rice grain).

Fig. 3.1 shows a section of the artificial DNA sequence used.

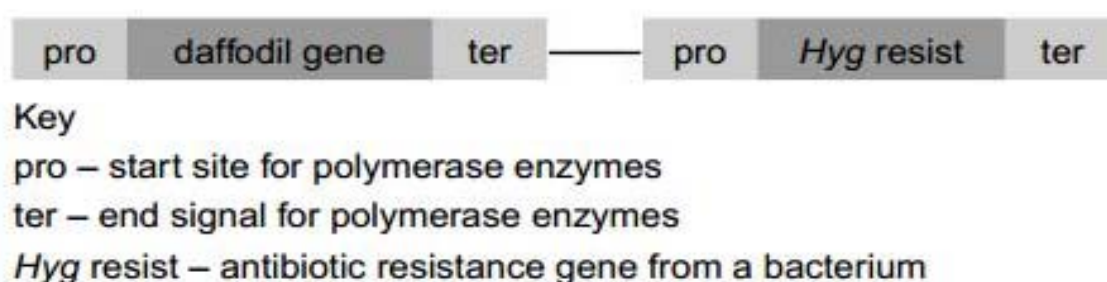


Fig. 3.1

Fig. 3.2 shows the main events in obtaining a transgenic plant.

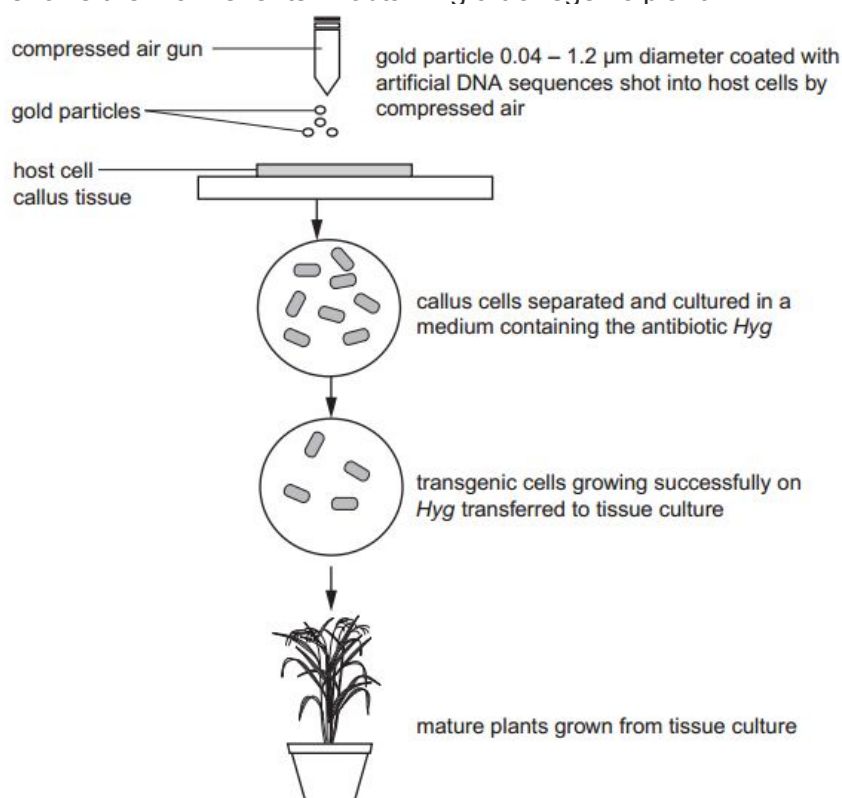


Fig. 3.2

- (a) With reference to Fig. 3.1 and Fig. 3.2,
- (i) explain what is meant by 'an artificial DNA sequence'. [1]
A DNA sequence that has been genetically altered by the addition of a gene or genes from another organism or from different organisms;;
- (ii) explain the significance of including the *Hyg* gene in the procedure. [2]
 1. ***Hyg-resist* gene confers resistance to the plant cells from the antibiotic Hyg;;**
 2. **Only transformed callus cells containing artificial DNA sequence with *Hyg* gene will survive in medium containing Hyg, selection of transformed calli possible;;**
- (iii) suggest a reason for the relatively low efficiency of obtaining transgenic plants by using an air gun (gene gun) to deliver artificial DNA sequences into the plant cells. [1]
 1. **Most DNA did not end up in the nucleus / end up in the cytosol and does not integrate into plant genome;;**
 2. **DNA containing daffodil gene is degraded by nucleases and does not integrate into plant genome;;**
 3. **DNA containing daffodil gene pass through the cells with the gold particles;; Any 1**

- (b) A separate investigation was carried out into the effect of different cytokinin concentrations on the development of rice callus cultures. The results obtained after six weeks are shown in Table 3.3.

	Cytokinin concentration/ %		
	0	20	40
Mean number of roots	20	0	0
Mean number of shoots	4	15	25
Size of shoots	Small	Large	Medium

Table 3.3

- (i) Briefly explain what is meant by 'callus culture'. [1]
Mass of, disorganised / undifferentiated / unspecialised, plant cells;
- (ii) With reference to data from Table 3.3, state the main conclusions that can be drawn from this investigation. [3]
- Higher concentrations of cytokinin promote greater number of shoots – increase of cytokinin concentration from 20% to 40% increased the mean number of shoots from 15 to 25 or an increase of over 66% (25-15/15);;**
 - Too high concentrations of cytokinin reduces the size of the shoots – increase of cytokinin concentration from 20% to 40% decreased the size of the shoots from large to medium/ 20% cytokinin concentration is the optimal concentration for size of shoots;;**
 - Roots developed only when there was no cytokinin / cytokinin inhibited root development – roots appeared only at 0% cytokinin concentrations, at 20 – 40% there was no root growth;;**

Each conclusion must be supported by data from Table 3.3

Many varieties of rice, *Oryza sativa*, will die quickly if they are totally submerged by flood water (submergence-intolerance).

A plant breeder and genetics scientist has just discovered a gene, *JJ8* that was found to be involved in tolerance of prolonged submergence in rice varieties. An allele, *JJ8-A*, was subsequently found only in submergence-tolerant rice.

Varieties that are submergence-intolerant either have a second allele, *JJ8-a*, or lack the gene altogether.

Typically, naturally submergence-intolerant plants respond to submerged conditions by elongating rapidly.

(c) Suggest a reason for plants elongating rapidly when they are submerged in water. [1]

1. To allow the leaves / stem to obtain oxygen for respiration;;
2. To allow leaves to obtain light for photosynthesis;;
3. AVP;; Any 1

The mean heights of three types of rice plants were compared before and after 10 days of submergence in water. The types of rice plants were:

- naturally submergence-intolerant plants, denoted by **I**
- transgenic submergence-intolerant plants expressing *JJ8-A*, denoted by **I⁺**
- naturally submergence-tolerant plants, denoted by **T**

The results are shown in Fig. 3.4.

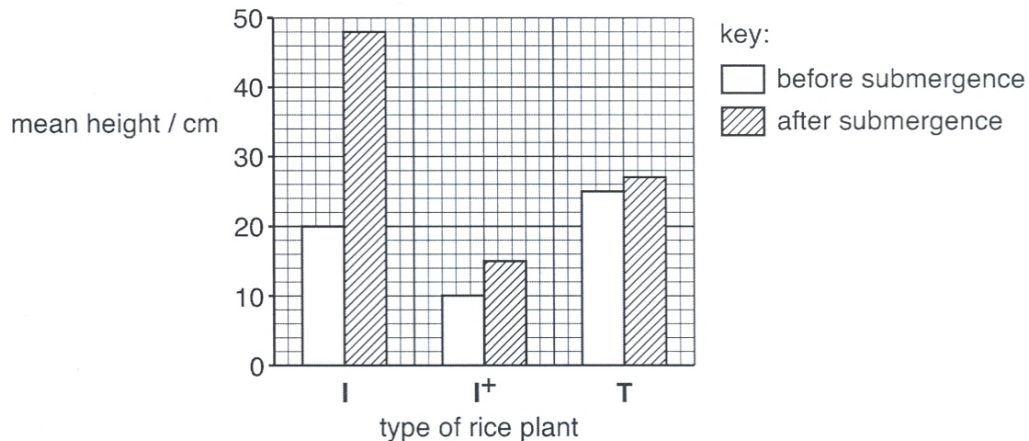


Fig.3.4

(d) With reference to Fig. 3.4, describe the differences in mean height before and after submergence in the three types of rice plants. [3]

1. **I⁺ extends x1.5 times / grows 5 cm compared to before submergence;;**
2. **T – only very slightly taller / insignificant change / grows by only 2 cm compared to before submergence;;**
3. **I more than doubled height / x2.4 times / grows by 28 cm compared to before submergence;;**

The allele *JJ8-A* could be introduced into a high-yielding variety of submergence-intolerant rice either by genetic engineering or by traditional selective breeding (crossing) of different rice plants.

(e) Suggest **two** advantages of using genetic engineering, rather than traditional selective breeding, to introduce the allele. [2]

1. **Only desired trait(s) are transferred;;**
2. **Transgenic organisms can be produced, with genes from other species;;**
3. **Shorter time taken to see results compared to traditional selective breeding;; R: 'shorter time' without elaboration**
4. **Does not dilute desirable traits of high yield variety;;**
5. **Does not dilute alleles of other useful genes;;**

[Total: 14]

4 Planning Question

An α -amylase obtained from a species of bacterium, *Bacillus subtilis*, hydrolyses amylose to glucose.

A colorimeter is a device that can be used to measure the concentration of a solute by measuring the absorbance of different wavelengths of light in a solution. Generally, a solution of higher concentration will absorb more light energy at a specific wavelength. Therefore, the higher the absorbance value, the higher the concentration of a solution. Although absorbance has no units, it is often recorded as absorbance units (A.U.)

Design an experiment to test the hypothesis that:

The rate of glucose produced is dependent on the concentration of α -amylase.

You must use:

- 5.0% stock α -amylase solution which can be diluted to give various solutions of amylase concentration
- 1% starch solution
- Benedict's reagent
- 5 x specimen tubes
- 6 x boiling tubes
- 6 x plastic vials for containing solution to be placed in colourimeter
- 2 x 5cm³ syringe
- Bunsen burner
- Colourimeter attached to data logger
- Thermostatically controlled water bath set at 37°C

Your plan should have a clear and helpful structure to include:

- a description of the method used including scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- correct use of technical and scientific terms,
- relevant risks and precautions taken.

[Total: 12]

Mark Scheme

- 1 Identification of variables – independent and dependent variables.
- 2 Identification of other variables to be kept constant (name 2 variables)
- 3 Theoretical consideration or rationale of the plan to justify the practical procedure.
 - prediction of likely experimental outcome
- 4 Theoretical consideration or rationale of the plan to justify the practical procedure.
 - explanation of likely outcome
- 5 Description of method used – using colourimeter to measure brick red precipitate obtained from Benedict's test
- 6 Dilution table to obtain 5 different concentrations of amylase (at regular intervals)
- 7 Set up of control (0 % amylase or boiled amylase)
- 8 Equilibrate amylase solution and glucose at 37°C for a fixed duration first (3-5 minutes).
- 9 Benedict's test set up - equal volumes of Benedict's reagent and sample; boiling water bath
- 10 Replicates and at least two repeats
- 11 Calculation of average in table
- 12 Layout of and graph with appropriate axes. rate of glucose production/s⁻¹ (y-axis) is plotted against amylase concentration/ % (x-axis)
Accepted units for rate of glucose produced- A.U./s or A.U./min
- 13 Safety precaution / risk hazard identified with explanation
- 14 The correct use of technical and scientific terms

[Total: 14, max = 12]

Variables

- The independent variable in this experiment is amylase concentration while the dependent variable is absorbance.

Other variables to be kept constant

- volume of Benedict's reagent used for Benedict's test
- volume of sample used for Benedict's test
- volume of amylase
- volume of starch
- constant temperature for equilibrating amylase and starch e.g. 37°C
- fixed reaction time between amylase and starch e.g. 5 minutes
- constant pH

Prediction of the likely outcome of the experiment

- As amylase concentration increases, amount of glucose increases
- (An increasing amount of glucose leads to a more intense brick red precipitate after carrying out Benedict's Test)
- A greater amount of brick-red precipitate gives rise to a higher absorbance value

Explain the likely outcome of the experiment

- As the concentration of amylase increases, the number of available active sites increases. Hence, frequency of successful collision between amylase and starch increases which leads to increased enzyme-substrate complexes formed and products/glucose formed. The rate of glucose formation increases.
- The higher the concentration of glucose, the greater the amount of brick-red precipitate produced after Benedict's test resulting in higher absorbance value of solution in colorimeter.

Method

- Intensity of brick-red precipitate formed after Benedict's test can be measured by using a colourimeter attached to datalogger to give an absorbance value.
- Rate of glucose production can be measured indirectly by dividing absorbance by the time taken (5min)

Procedure(includes safety and reliability of results)

1. Prepare 20 cm³ each of 1.0%, 2.0%, 3.0%, 4.0%, 5.0% α-amylase solution by diluting the 5.0% stock α-amylase solution as indicated in the table below and **a control of 0.0% α-amylase solution/ boiled 5.0% α-amylase solution.**

Specimen tubes	Concentration of α-amylase solution / %	Volume of 5% α-amylase / cm ³	Volume of distilled water / cm ³
A	1.0	4.0	16.0
B	2.0	8.0	12.0
C	3.0	12.0	8.0
D	4.0	16.0	4.0
E	5.0	20.0	0.0
Control	0.0	0.0	20.0

2. Set up a **thermostatically-controlled water bath at 37°C.**
3. Set up Tube Y containing 2cm³ of 1.0 mol dm⁻³ starch solution.
4. Transfer 2 cm³ of mixture from Tube A into a new boiling tube labelled A1.
5. **Equilibrate** Tube Y and Tube A1 in the water bath maintained at 37°C for 5 minutes.
6. After 5 minutes is over, pour the reaction mixture in Tube Y into Tube A1.
7. Use a stop watch to time for 5 minutes for the reaction between starch and amylase to proceed.
8. Add an **equal volume** (4 cm³) of **Benedict's solution** to the same Tube A1.
9. Place Tube A1 into a **boiling water bath** and use a stopwatch to time for two minutes.
10. Remove Tube A1 from water bath.
11. Place 1 cm³ of solution into a vial to measure for absorbance using a colorimeter.
12. **Repeat step 3-11 to obtain 2 more readings.**
13. **Repeat step 3-12 using 2.0, 3.0, 4.0 and 5.0% α-amylase and 0.0% α-amylase (for control).**
14. **Repeat the entire experiment twice.**
15. Plot a graph of rate of production of glucose/ A.U/s against concentration of α-amylase / %.

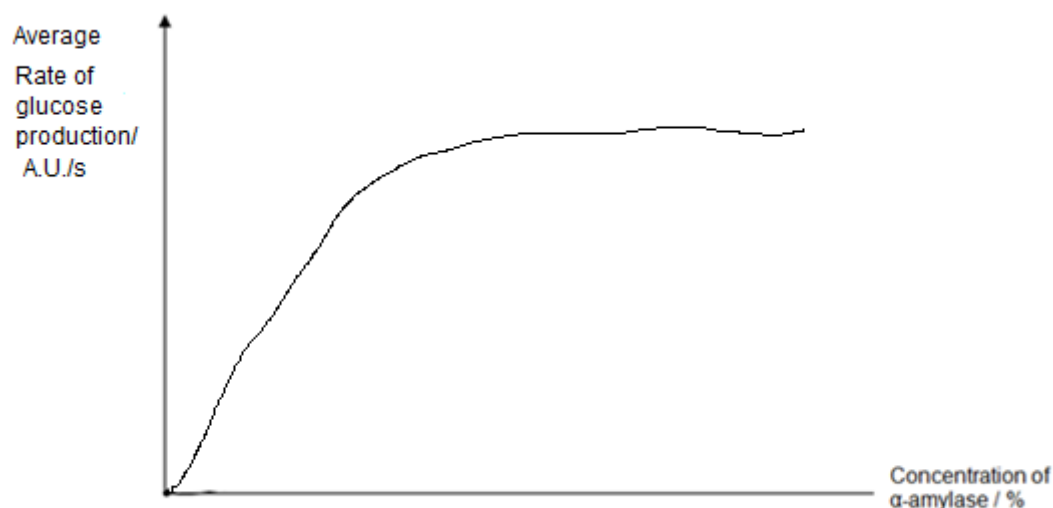
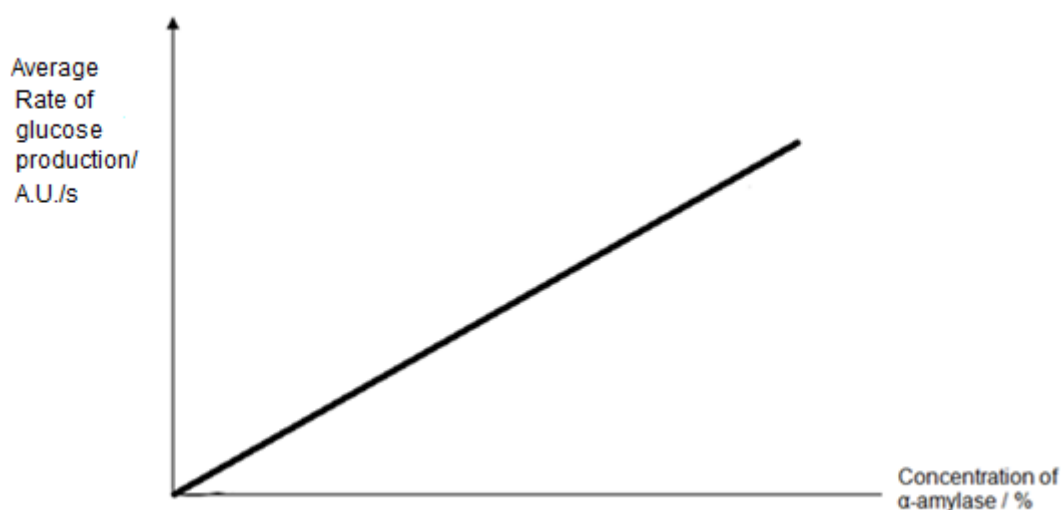
Possible additional steps:

16. Repeat step 8- 11 using 0.5%, 1.0%, 1.5% and 2.0% glucose concentration to obtain absorbance value using a colorimeter.
17. Plot a graph of absorbance against glucose concentration.
18. Using the graph in step 16, the amount of glucose produced by different concentrations of α-amylase can be read off.
19. Plot a graph of rate of production of glucose/ %/s against concentration of α-amylase / %.

Table showing rate of glucose/A.U./s at varying concentration of α -amylase/%

Concentration of α -amylase / %	Absorbance / A.U.				Rate of glucose production/ A.U/s
	Reading 1	Reading 2	Reading 3	Average	
0.0					
1.0					
2.0					
3.0					
4.0					
5.0					

Table showing rate of glucose/ A.U/s at varying concentration of α -amylase/%



Risk Assessment

- Remove the thermometer from the water bath immediately after use and place it in a safe place e.g. drawer, to prevent it from rolling onto the floor.
- Handle fragile objects like test-tubes with care. Arrange test tubes neatly in the test-tube rack, ensuring that they are placed where they cannot be knocked over or rolled off the bench.

Free-response question

Write your answers to this question on the separate paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections (a), (b) etc., as indicated in the question.

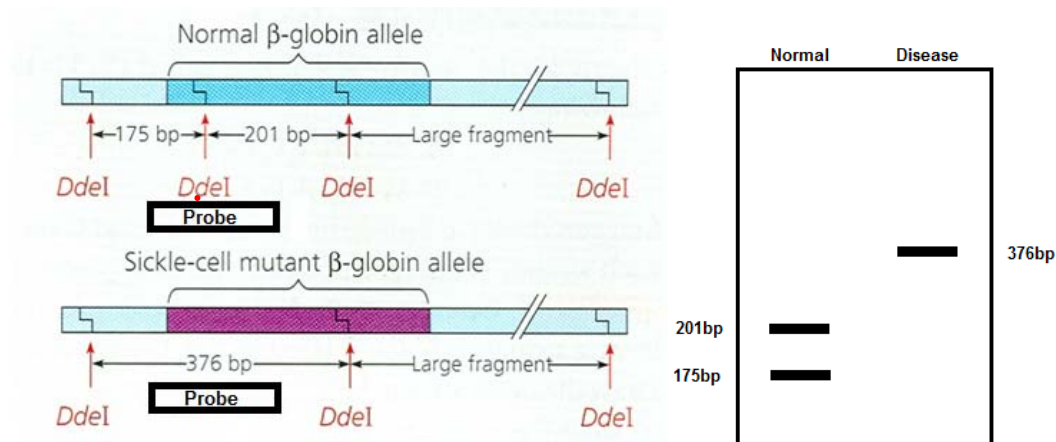
5 (a) Explain how gel electrophoresis is able to separate fragments of DNA. [6]

1. Gel electrophoresis is a procedure for separating a mixture of DNA fragments through an agarose gel in an electric field;;
2. An agarose gel with wells at one end is placed in a buffer containing salts/ions that will conduct electricity;;
3. DNA samples are mixed with loading dye/bromophenol blue to help DNA to sink to the bottom of the well/help in monitoring the progress of the electrophoresis before they are loaded in the wells near the cathode/negatively-charged electrode;;
4. When the current is switched on, negatively-charged DNA fragments will migrate towards the anode/positively-charged electrode;;
5. Rate of movement is inversely proportional to length/size/molecular weight of the DNA fragment;;
6. Longer length of DNA fragments encounter more resistance, travel more slowly through the pores in the gel / travel shorter distance through the gel from the well and will be nearest to the wells;; (or vice versa)

(b) Explain how RFLP analysis has helped in the direct detection of sickle-cell anaemia. [6]

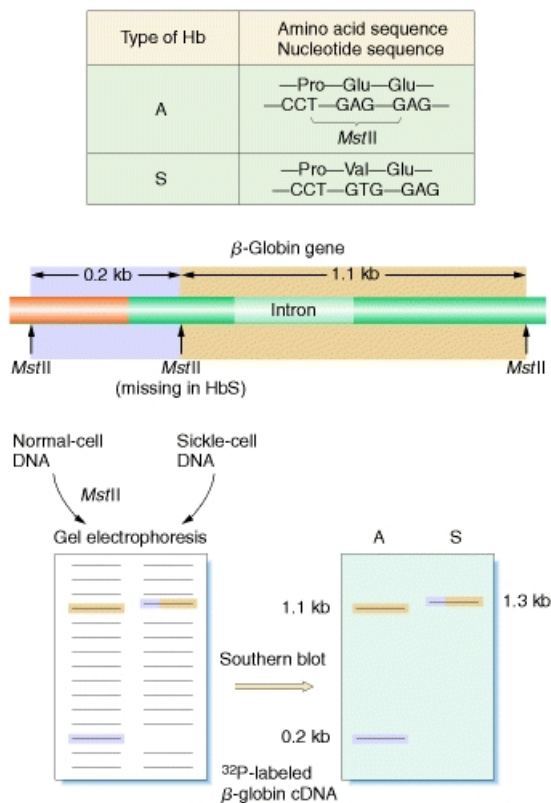
1. Disease-causing mutation is a single base-pair substitution which occurs within the β -globin gene;;
2. Such a mutation occurs at a restriction site (5'CTGAG3') which is recognised by the restriction enzyme DdeI;;
3. Therefore, compared to the normal allele (Hb^A), disease allele (Hb^S) has one less restriction site / restriction site was eliminated due to the mutation;;
4. When both alleles HbA and HbS are digested using DdeI, restriction fragments of different size are produced;;
5. These fragments can be separated using gel electrophoresis: Rate of migration is inversely proportional to the length/size/molecular weight of DNA fragment;;
6. Southern blot can then be carried out using a probe complementary to part of the β -globin region;;
7. The presence of a disease allele in an individual can then be seen from characteristic banding pattern in Southern blots;;
ref different band sizes for normal vs sickle-cell anaemia

(Normal individuals ($Hb^A Hb^A$) will have 1 band at 175kb and 1 band at 201kb while individuals with sickle-cell anaemia ($Hb^S Hb^S$) will have 1 thick band at 376kb)



8. A sketch of the above diagrams will be useful in showing the positions of the probes used and explaining the expected banding pattern;;

Note: Ref. to *MstII* can also be accepted:



(c) Explain the benefits of the human genome project. [8]

1. The HGP allows for the discovery of genes/alleles/polymorphisms that can be used for genetic testing and DNA forensics;;

Genetic Testing

2. This allows for more accurate diagnosis of diseases and earlier detection of genetic dispositions to diseases (e.g. breast cancer, cystic fibrosis, Alzheimer's disease);;
3. It also allows scientists to design drugs to target a specific gene/protein (i.e. rational drug design);;

DNA forensics [max 2]

4. identify potential suspects whose DNA may match evidence left at crime scene, or clear innocent people of wrongdoings;;
5. establish paternity and other family relationships;;
6. match organ donors with recipients in transplant programs; the more genetically similar the recipient is to the donor, the greater the compatibility of the donor organ;;

Gene Therapy

7. The identification of genes associated with human genetic diseases and a better understanding of the genetic basis of disease may increase the potential to treat genetic and acquired diseases by gene therapy;;
8. Gene therapy allows defective genes to be either replaced by normal functional genes or supplemented with other genes to prevent a disease taking its course;;

Pharmacogenomics [max 2]

9. The HGP allows scientists/doctors to know which genes/alleles affect an individual's response to a drug and how these genetic differences affect the way the individual reacts to a drug;;
10. This may eventually allow doctors to prescribe the right/effective drug in the right dosage for each individual;;
11. It is also theoretically possible to design drugs based on molecular information that would work better and cause fewer side effects;;
12. Many customised drugs work by targeting specific proteins within our body. Knowledge of the gene sequence that codes for these specific proteins allows us to deduce 3D conformation of these proteins and therefore design drugs that are would target these proteins;;

Note: award once for Pt 3 or Pt 12

[Total: 20]